FABER CASTELL
Date
Page No.

1. Introduction

E-Commerce platforms process many transactions daily. Understanding transaction-amount distributions helps analysts describe typical purchases, identify unusual amounts, estimate probabilities, and support business decisions.

2. objective

- Study probability distributions, Q-Q plots, transformations, Probability functions, and Z-scores using Python.

3. Dataset and assumptions

- The brief specifies transaction ID, customer ID, amount, date, count, region, and status. No official dataset file accompanied the brief, so the workbook contains 500 rows of synthetic demonstration data. Replace it with the course dataset if available.

4. Theoretical foundation

4.1] Statistical distributions

- A statistical distribution describes how values or outcomes are spread and the probability associated with them. Examples include normal, binomial, Poisson, log-normal, and Power-law distributions.

4.2) Q-Q Plot

- A quantile-quantile (Q-Q) plot compares sample quantiles with theoretical quantiles. If points lie approximately on a straight line, the sample may

be consistent with the selected distribution. Curvature or tail departures indicate a mismatch. A Q-Q plot is a visual diagnostic, not a complete proof.

4.3] Discrete vs Continuous distributions

- A discrete distribution assigns probabilities to countable outcomes, such as 0, 1, 2, ... transactions. A continuous distribution models measurements over a range, such as transactions amounts. For a continuous variable, probability is represented by area under a pdf density curve; the probability at one exact value is zero.

4.4] Bernoulli distribution

- A Bernoulli random variable has two outcomes. Commonly success (1) with probability p and failure (0) with probability 1 - p. For this project, a transaction status can be coded as success = 1 and fail = 0.

4.5] Bernoulli distribution

- A binomial variable counts successes in n independent Bernoulli trials with a constant success probability p. Its probability mass function is $P(X = k) = C(n, k) \cdot p^k (1-p)^{(n-k)}$. It can model the number of successful transactions in a fixed group, if assumptions are reasonable

4.6] Log-normal distribution

- A positive random variable x is log-normally distributed when ln(x) follows a normal distribution

It is often right-skewed and can describe purchase amounts where many values are moderate but a few are large. It only supports positive values.

4.7) Power-law distribution

- A power-law relationship has a tail that decreases approximately as a power of the value, often written $p(x)$ proportional to $x^{(-\alpha)}$ for an appropriate range. It is used to describe heavy-tailed phenomena. A convincing power-law claim requires tail-threshold selection and comparison with alternatives; a simple fit alone is not enough.

4.8) Box-Cox transformation

- For positive x, the Box-Cox transformations is $(x^{\lambda}-1)\lambda$ when $\lambda \neq 0$ and $\ln(x)$ when $\lambda = 0$. The parameter λ is estimated from data. The transformation can reduce skewness and make variance more stable.

4.9) Poisson distribution

- A Poisson variable counts events in a fixed interval when events occur at a stable average rate and independently. $P(X=k) = e^{(-\lambda)} \lambda^{k}/k!$, where λ is the expected count in the interval. ~~Ex~~ Example: the number of transactions received per day, with λ estimated by the average daily count.

FABER-CASTELL
Date
Page No.

4.10] Z-Score Probability

- A z-score measures how many standard deviations an observation lies from the mean: $z = (x - \mu)/\sigma$ (or sample estimates when μ and σ are unknown). Under a normal model, a probability above a threshold a is $P(X \geq a) = 1 - \Phi((a - \mu)/\sigma)$. This normal-model probability should be used cautiously when amounts are strongly skewed.

4.11] PDF and CDF

- A probability density function (PDF) describes density across a continuous range; probabilities are areas under it. A cumulative distribution function (CDF), $F(x) = P(X \leq x)$, gives the probability a value is at most x. The CDF is non-decreasing and ranges from 0 to 1.

5. Practical methodology.

1. Inspect data quality and types with Pandas.
2. Model success/failure with Bernoulli and grouped successes with Binomial.
3. Count transactions per day and estimate Poisson λ.
4. Fit log-normal and exploratory Power-law models to amounts.
5. Generate a normal Q-Q plot and apply Box-Cox

transformation.

6. Calculate Z-scores and plot PDF/CDF; compute observed share above ₦5,000.

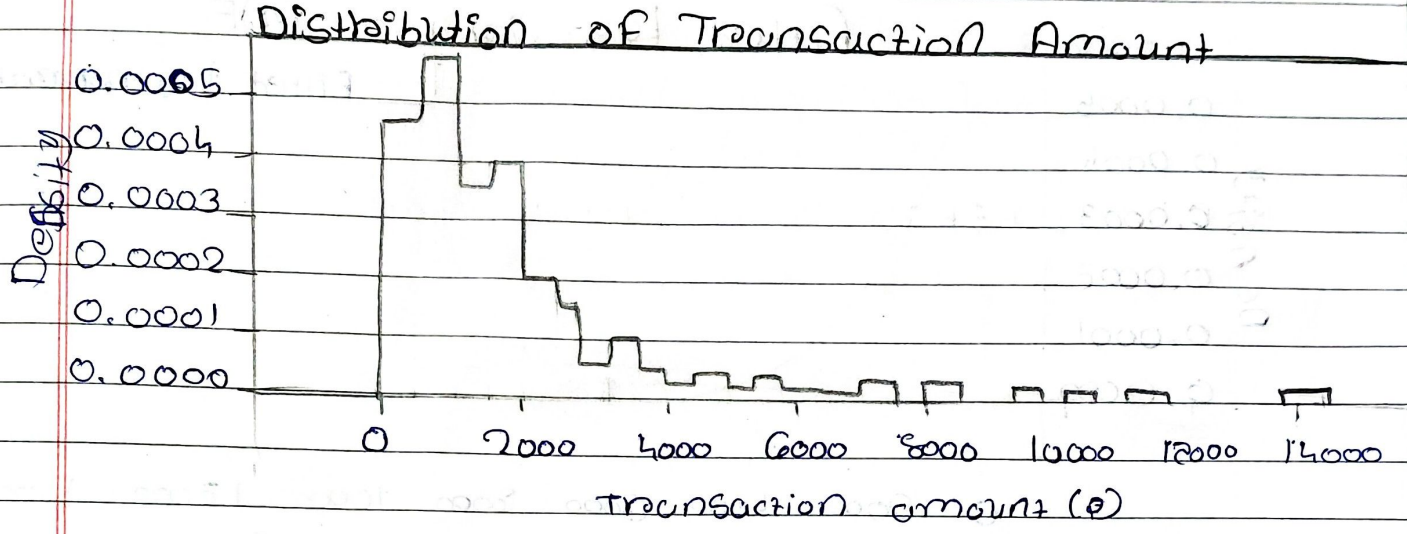
6. Demonstration results

Metric	Value
Number of transactions	500
mean transaction amount	₦1,666.90
median transaction amount	₦1,203.28
Sample standard deviation	₦1,664.13
Skewness	3.332
observed share above ₦6,000	4.6%
success rate	92.8%
Mean transactions per calendar day	6.56%
Estimated Box-Cox λ	-0.039

Mean = ₦1,666.90; median = ₦1,203.28; skewness = 3.33. The sample is right-skewed. The observed share above ₦5,000 is 4.6%. these are synthetic-data results.

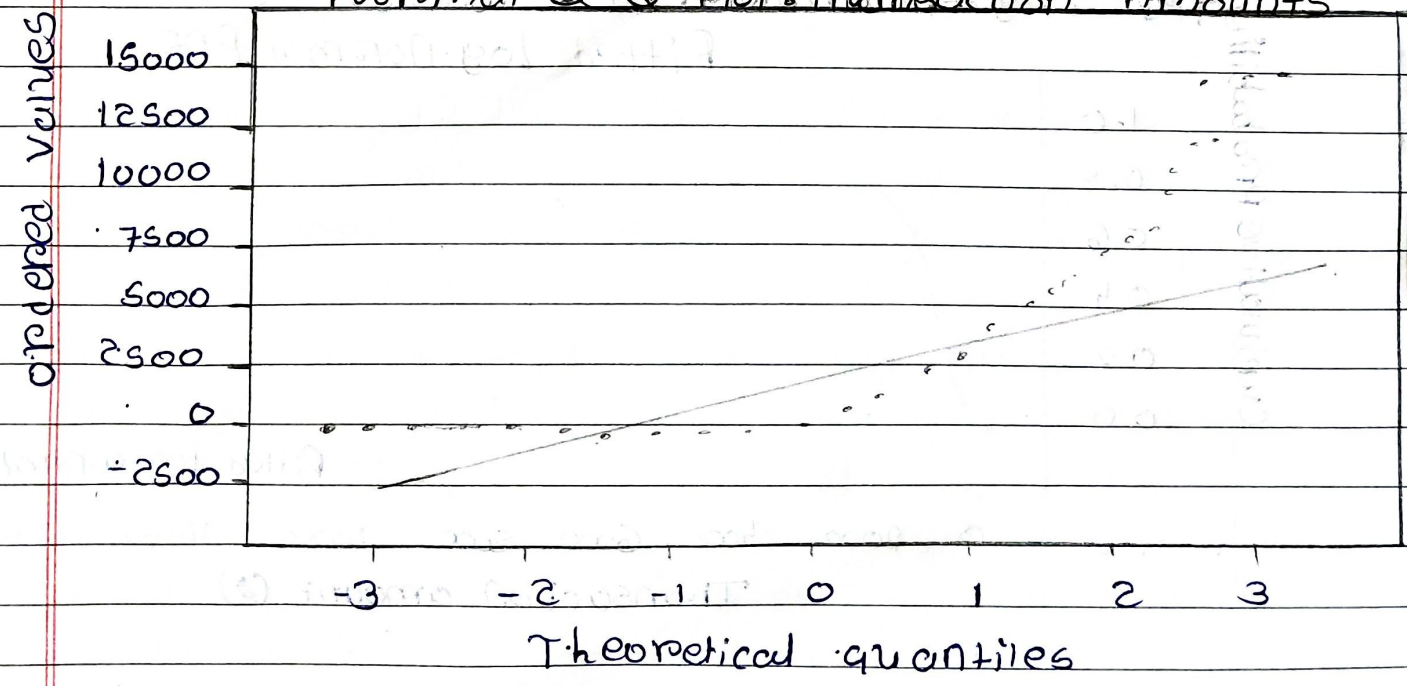
FABER CASTELL
Date
Page No.

7.1 Histogram



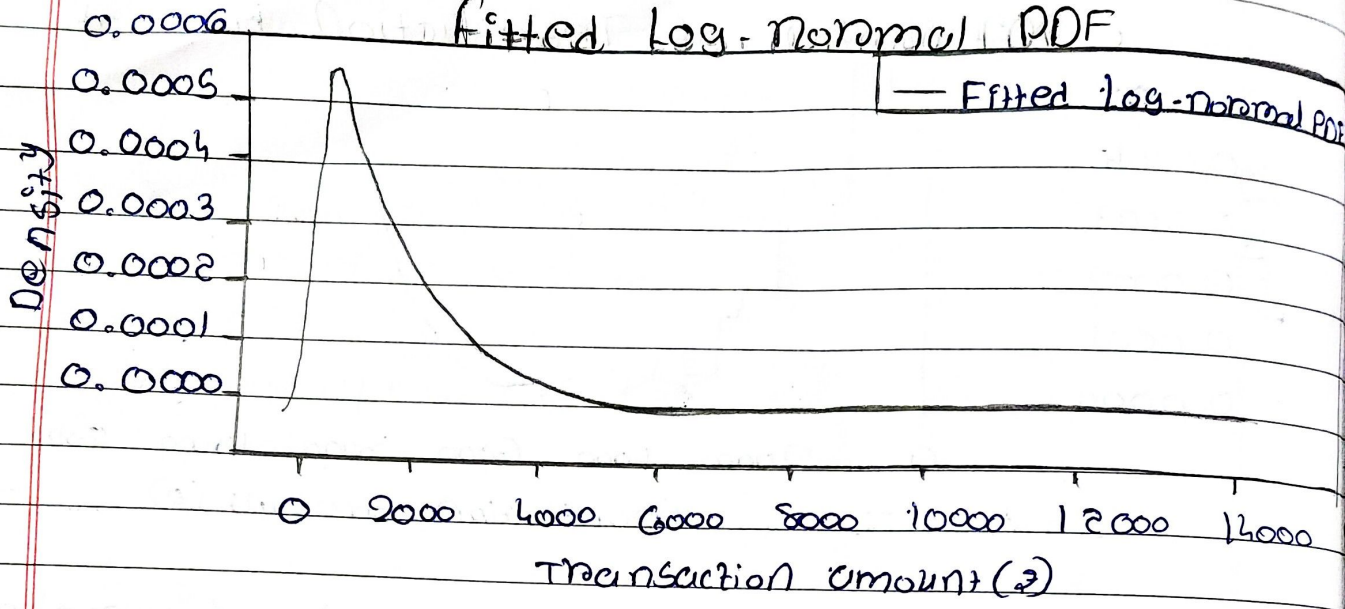
A long right tails suggests a small number of large purchases

7.2) Normal Q-Q Plot.



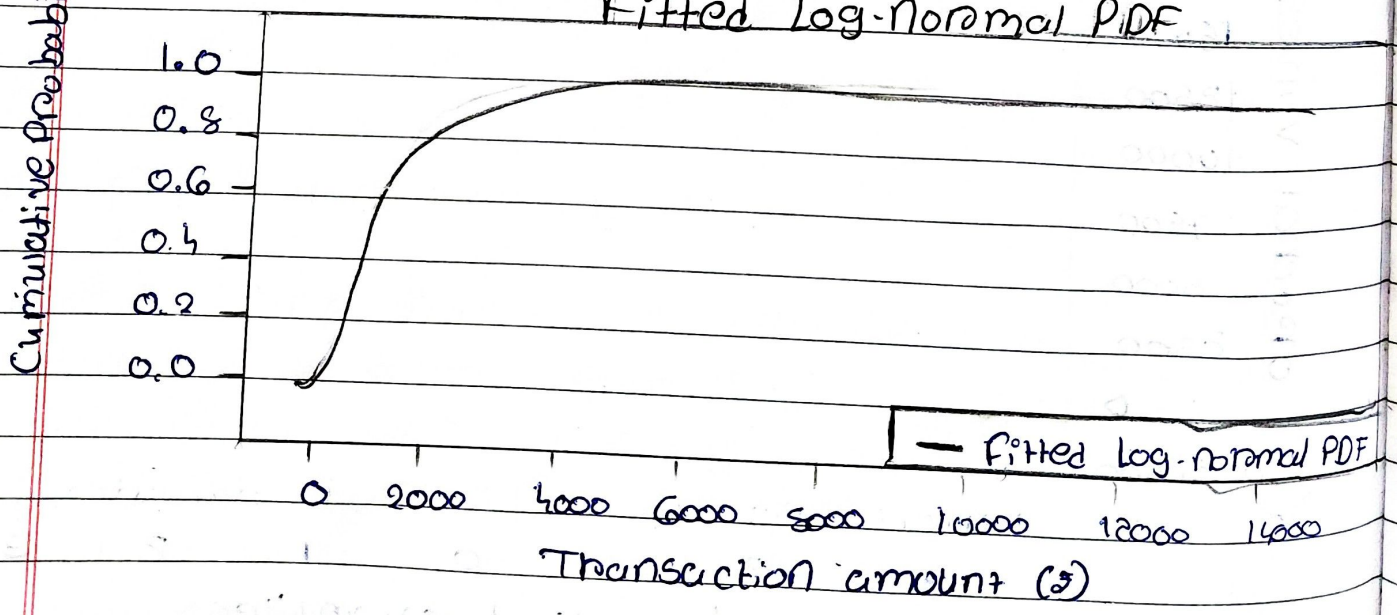
- Tail departures from the line indicate mismatch with a normal model.

7.3 Log-normal PDF



- Density values are not point probabilities; areas over intervals represent probability.

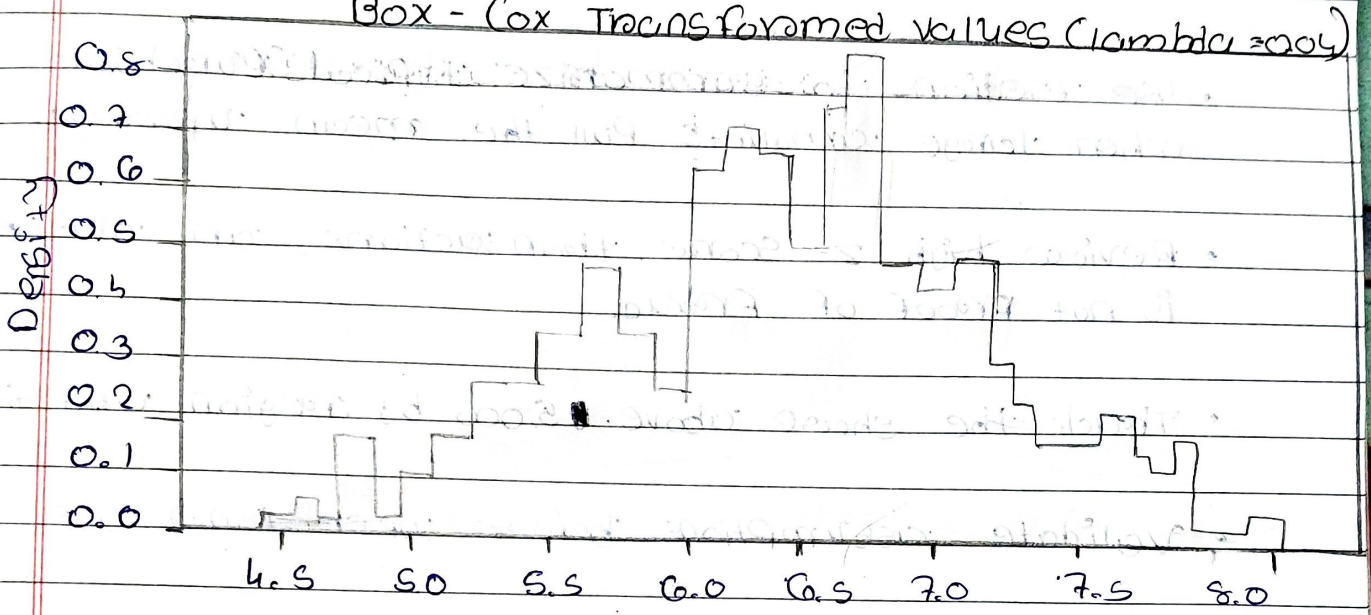
7.4) Log-normal PDF



- The CDF estimates: $P(X \leq x)$; 1 - CDF estimates $P(X > x)$.

FABER-CASTELL
Date
Page No.

7.5) Box - Cox histogram



- Compare ~~Skewness~~ skewness before and after transformation; normality is not guaranteed.

8. Z - Scores and threshold

∴ z = (amount - sample mean) / sample standard deviation. Positive values are above the mean; negative values are below. The observed share above ~~Shape~~ ■ 5,000 is 4.6%. Because the data are skewed, a normal - model tail probability should be used cautiously.

9. Distribution selection and business insights

- A log-normal model is a reasonable first candidate for positive, right - skewed amounts, validate it using diagnostics and comparison with alternatives. Bernoulli suits binary status, Binomial counts successes in fixed, and Poisson is a candida

FABER-CASTELL
Date
Page No.

for daily counts if assumptions hold.

- Use median to summarize typical purchases when large amounts pull the mean upward.
- Review high z-score transactions; an outlier is not proof of fraud.
- Track the share above ₹5,000 by region and time.
- Validate assumptions before operational use.

10. Limitations and Conclusion

The data are synthetic, so results do not describe real customers. Re-run the analysis on the official dataset. This project demonstrates distribution fitting, Q-Q diagnostics, Box-Cox transformation, z-scores, and PDF / CDF interpretation.

11. References

- Scipy: https://docs.scipy.org/doc/scipy/reference/stats.html
- Numpy: https://numpy.org/doc/
- Pandas: https://pandas.pydata.org/docs/
- Matplotlib: https://matplotlib.org/stable/
- Course brief: Spread Locator: A Statistical Distribution Analysis Model.